ActInf GuestStream 108.1 ~ Next-Generation AI for Emergent Semantic Co

GuestStream | Next-Generation AI for Emergent Semantic Communications (Christo Thomas) | 1:08:51

Hello, welcome. It is May 22nd, 2025. We are in Active Inference Guest Stream 108.1 discussing

Next Generation Artificial Intelligence for Emergent Semantic Communications with Christo

K. Thomas. So Christo, thank you very much for joining. Looking forward to hearing about the work and also to

everyone's comments and questions from the live chat. Thank you. To you.

Thanks, Daniel, for the introduction. Yeah. And thank you for inviting me for this talk.

So I'll be talking about Next Generation Artificial Intelligence for Emergent Semantic Communication

and also I would like to express my gratitude to my collaborators, my postdoc advisor, Dr.

Valit Saad and PhD student in our group, Omar Hashash. Yeah. So I would also like to advertise a bit on

the upcoming research group I will be establishing as a tenure track assistant professor at

Worcester Polytechnic Institute in Massachusetts. So therein, my focus will be to work at the

intersection of Next Generation Artificial Intelligence and Wireless Systems with specific focus on

resilient and interoperable semantic communication and integrated sensing and communication frameworks,

explainable and generalizable AI for Next Generation Wireless Systems,

foundation models for networking applications and developing strong mathematical foundations of

cognitive and reasoning driven AI architectures. So yeah, so this is an outline of the talk.

So first I will go over a motivation of Next Generation AI Native Wireless Networks.

Next Generation AI Native Language Networks. So then further I will talk about two of my recent research work on semantic communication

and further I will provide some future perspectives which involve emerging language design and using active inference.

So yeah, so if you look at the wireless evolution right now we are at the fifth generation of wireless systems,

which promised apart from you know cellular enhancing the cellular communication aspects like massive machine type of communication or ultra reliable low latency communication.

However, 5G could not meet all the promises it actually made.

So yeah, as like the you know saying goes usually the old generation or numbered generations are not very successful at their what they promise. So if that is true,

so usually you know the even number of generation so next one will be the 6G so we can expect more exciting developments in

6th generation of wireless systems. So in 6G so we are not okay as we move towards making our environments more smarter through you know making the network intelligent

and be able to control the actions of distant autonomous agents in the environment.

So in order to support those applications 6G brings in a lot of different more distant set of challenges like you know trustworthiness, security and resilience of underlying AI models,

immersivity, sensing, AI nativity and sustainable computing.

So here AI is going to play a crucial role in the you know design management or maintenance of different network and device functions.

So yeah, so such a transition from the fifth generation to AI native sixth generation is expected as you move to sixth generation of technologies.

So here the question is why AI native because it can brings in the aspects of you know adaptability to

dynamically varying wireless environment or tasks. It can handle non-linear signals.

It can also bring in a lot of resource efficiency compared to classical systems. So these are some of the advantages that AI nativity provides.

However, if you look at the state of the art of AI native networks. So currently the literature is predominantly data driven AI algorithms like you know

so yeah meta learning or autoencodes, CNNs and right now you know there is a huge high point using large language models for

wireless network optimization. However, these data driven machine learning algorithms

are plagued by the following disadvantages like large data size which may not be always available readily in a wireless system.

Second is their limited generalizability to changing wireless environments or tasks.

So yeah associated with the limited generalization is the aspect that we need frequent retraining of these machine learning models.

So we need to look at the data driven machine learning algorithms which means we yeah if we have to build you know communication I mean real time communication and control such data driven machine learning algorithms are not suitable.

So another aspect is they act like black box models so they lack interpretability and then they require huge computing power.

So you know unsustainable computing is another disadvantage of these machine learning algorithms.

So as we transition from the current AI native to next generation AI native we must advance artificial intelligence together with the wireless network evolution.

So that's the focus of a few of our research papers, the current vision papers that we

uh some I mean that got accepted in the last couple of years.

Uh so one paper in one in one paper we actually discussed how how we can actually use next generation machine learning algorithms like causal reasoning for

um diverse um or cross-layer network optimization tasks

and another paper that actually got accepted at the IEEE proceedings of IEEE we actually looked at a rigorous artificial general intelligence

based vision um which I will summarize as follows.

So yeah so next generation wireless systems are not just about communication so this those will also involve lot of sensing.

So sensing act like an eye for the network so using the sensed information that contains the states of different autonomous agents or cyber physical systems in the environment.

The network should then act like a telecom brain so where telecom brain is characterized by distinct um capabilities such as perception which means the ability to

generate abstract representations about different semantic concepts present in the data.

Then using these abstractions to perform or compute or learn a world model that actually allows it to understand the

cause and effect relations among different physical processes in the world.

So this world model will allow it to perform

or predictions about future networks or environment states and then use that predictions to basically perform action planning on the network.

So here action planning refers to not just network actions like you know resource allocation or beamforming but also

designing optimal control actions for the autonomous agents in the environment.

So as to you know guide or you know control them um in the way as that the network intend it to perform.

So that's the idea of the you know telecom brain principle um that we actually envision.

So along this direction uh one aspect is the perception or the semant extracting semantics from the data.

So further in this talk i'll be talking about you know how we can uh extracting semantics and then

communicating the semantics how it can actually enhance the resource efficiency of next generation

wireless system. So that will be the focus of the um next few slides.

So to introduce semantic communication so if you look at the classical communication view uh so therein the

objective was to you know ensure that the messages are recovered at the receiver side with the arbitrary low probability of error as possible.

So that is the technical problem. So while doing so we have so far ignored the semantic aspect which means

what meaning do these transmitted symbols convey and the effectiveness problem which means how effectively

does the received meaning affect conduct in the desired way at the receiver side. So yeah as Shannon has also

mentioned the semantic aspects are irrelevant to the engineering problem. However when we move towards

future networks wherein you need uh we have we have a lot of stringent communication requirements as well as a lot of other

you know intelligent requirements like you know trustworthiness or immersivity as i discussed before

uh we cannot continue to rely on just the technical problem. So in order to make the communication resources more efficient

uh we must transition to a novel paradigm called semantic communication. So in semantic communication the

objective is to transmit the meaning behind the messages. So here meaning actually means only the information that is relevant to the end user. So communicate this information

information relevant information by building a semantic language between the transmitter and receiver. We call the transmitter to be a teacher and apprentice to a receiver to be an apprentice.

So yeah so by creating a machine language uh the objective is to ensure that receiver is able to regenerate the meaning uh with high fidelity as possible.

So here the question is so what is this semantic semantics actually contained in the uh data uh so yeah so if you look at this state of the art one traditional framework is

this joint source channel coding framework framework so where so in traditional system we used to

separate the source encoding and the channel encoding part but in joint source channel coding framework we
actually combine the source and channel encoding to one module and in deep joint source channel coding we use a machine
learning and based encoder and decoder and then train them in an end-to-end fashion with the goal of minimizing
certain semantic loss function. So by training such machine learning models we achieve we ensure that transmitter
extract only the relevant information or relevant features from the data and communicate that information only.
So that leads to better compression uh in terms of what is being communicated. However our vision is that
semantic communication is not just about you know compressing but better compressing the data. So instead
so if you can install machine reasoning at the you know transmitter and receiver side let's say
receiver has better rig i mean uh you know prediction capabilities or uh ability to you know generate uh
missing information so which can be achieved through you know instilling uh advanced machine learning framework
so then we can achieve more minimalism in communication compared to just doing deep joint source channel coding
and then if we can instill these generalizable machine learning algorithms that actually ensure that
these we don't need to retrain the system as tasks or the wireless environment changes so that is also very
crucial for enabling real-time communication and control and another goal is to ensure high fidelity at the
receiver side. So in my previous work have so far tackled these three challenges which i will talk about in detail and
but there are also other open challenges like how to ensure interoperable multi-user semantic system
when each of the users speak different semantic languages or how to build a resilient semantic
communication system so resilience against any network disruptions that could be due to
time varying channel conditions or cyber attacks and then how to integrate these SC principles into the
current 3GPP standards without introducing significant interoperability and standardization challenges
so those are some open problems that i am currently looking at uh uh yeah so let me first talk about uh my
first work on semantics that got published in iEEE transactions on wireless communication so this is
about building a reasoning foundation for emergent semantic communication so in particular i consider a
a two two two user semantic communication system so wherein the transmitter uh observes the world state
and extracts the entities present in the data and then using the entities extracted it computes a semantic
state description which i will talk about in later a detail later which is okay denoted by isert belonging to
syntactic space and then the syntactic space i mean isert is encoded uh to a representation space and then communicated at the receiver
receiver extracts a semantic state using the extracted semantic state receiver performs certain reasoning
which is denoted by phi of isert hat and this recent output is used to perform certain actions on the world state
so here there are two problems that i consider so one is the reasoning problem so which involves how to

learn the semantics present in the data or how to define this semantics

and at the receiver after reconstructing the semantic state how the receiver can perform detective reasoning and given this detective reasoning how the no transmitter can

uh better encode this encode is another problem which is the emergent language problem so wherein

at the start of communication transmitter is not aware of how there is what are the semantics that receiver is interested in

and the receiver is not aware of how the transmitter is in which language that the transmitter will be going to encode semantics

so learning this encoder and decoder those are which are formulated as you know probabilist probability distribution

given as shown here pi of u given isert and pi of isert given x comma c c here actually corresponds to the communication context

so that will be the history of semantics that communicated in the previous time instance so that is the emergent language

problem yeah so for so so so for the reasoning framework uh we use uh causality from machine learning so where causality is founded on three different principles one is the causal discovery

so given the uh you know entities present in the data so here we call these entities as semantic content elements so only those entities that are

really relevant for the end user task so those are the semantic content elements so we learn the

causal effect relations between these entities as a directed acyclic graph as is shown here so from causality literature this can be called as the structural

causal model and this structural causal models apart from these semantic content elements it could also involve certain exogenous

random variables that are actually uh unknown or hidden from the um uh hidden from us yeah so that's a causal discovery phase so second phase is the causal representation learning so so when we are given a large dimensional observation maybe it could be larger dimensional antenna measurements

so we it may not be immediately available i mean immediately be known to us what are the causal

cause and effect variables so we need to first convert it to a lower dimensional latent representation

involving just the causal variables so that is the causal representation learning part so do you so during

this causal representation learning phase we can actually filter out the relevant and irrelevant variables from the

um features extracted from the data then finally we have the causal inference phase so wherein if we are given this

causal graph we can perform intervention that actually means we can manipulate the values of certain endogenous variables

here then analyze what is the impact that cause on their effect variables and then we can also perform

counterfactuals that means so given we have observed certain set of entities as is shown here so what would the world state would be if these

you know let's say s4 if you want if you want if you want to imagine a different outcome for s4 so what what

interventions on these causes would have caused that so such imaginary words we can assume using the framework of

counterfactuals so if we can actually okay extract this causal graph and then perform training on this

interventional and counterfactual distribution then we can actually attain a level of distribution

generalization so that's the advantage of bringing in causality and another thing is the explainability

aspect that so given this causal graph we can actually tell like you know if you have a causal graph

among uh distant environment variables then resource allocation communication and computing resource allocation

variables and quality of service values then we can actually assess if the performance falls

below certain threshold so what are the underlying causes that actually lead led to that particular

behavior so that that's where this explainability aspect is coming by using the framework of causality

so now okay so using the causal framework we learn the causal graph from the data so learning the causal

graph we formulate it as a factorized posterior distribution where each factor will be the probability of

each of these node variables given the its parent so probability of s1 given probability of so

probability s1 given s0 where s0 is the parent of s1 so that probability distribution we can use we

we learn using a recently popular a method called generative flow networks wherein we start with an

unconnected set of entities and iteratively learn each edge starting from this unconnected set of entities

and then at the receiver side after receiving this causal state receiver actually combines these smaller

states and then perform a logical reasoning where this logical reasoning is performed by evaluating certain

logical formula which is composed of these causal states that are connected by certain logical connective so

it's like evaluating so okay if this happened uh if the set of okay if these are the set of um i mean if this is the

causal state that the transmitter has observed okay then what would happen in the future okay yeah so those kind of

reasoning or inferences can be um performed using this logical inference uh framework so for that we use

uh we use uh we use uh we use uh existing uh

existing neurosymbolic ai method called logical neural network that actually converts this logical formula to a

neural network uh yeah so that's how we actually perform the logical reasoning at the receiver side

so now uh yeah i will

uh yeah i will consider the i will discuss the emergent language problem so so in the emergent language

problem the teacher's objective is to ensure that it communicates very less semantic information as

feasible uh so trans yeah the transfer is kind of uh uh yeah so that's the goal of transmitter uh so which is

formulated as negative of average semantic information uh at the transmitter side so semantic information

is that conveyed by the transmit semantic symbol u about the causal state uh is said

given the policy given the uh encoder policy as well as the history of

causal states at the receiver side receiver's objective is to maximize the

reconstructed semantic information uh so after receiving you how much information that can provide about the causal state is that at t

which is the reconstructed one given the policy of transmitter as well as the policy of the receiver and the history of causal states reconstructed which are the yeah then there are certain constraints like you know communication cost

plus minus log pi ld which is an apprentice surprise factor so which actually tells us how much the apprentice is surprised after

the receiving certain semantic information so yeah so here given this contrasting objectives of teacher and

apprentice we can formulate this as a two-player signaling game and the nash equilibrium of this two-player signaling game is

uh can be formulated as is shown here so yeah like apprentice responding to the uh given the

uh you know teachers optimal encoder policies what's the best response that actually you know apprentice can

uh provide that's that that's what this expression actually tells us so we actually solves this through

a since given so we actually formulated novel semantic information measures for the transmit side and receiver side

but we observed that those are non-convex functions so we sold them through an alternating optimization procedure where

alternating optimization is performed for some surrogate functions that are convex so yeah so this actually

evolves this game evolves through a two-way conversation between teacher and the apprentice wherein apprentice

asks certain questions about the data set data and teacher responds by providing an encoded version of the

information so that's how this game actually proceeds so as i mentioned here we cannot continue to rely on

the traditional mutual information measures since we need to quantify uh semantic information here so as i mentioned here

how i formulate semantic information is that so semantic information is formulated as the information contained in

the logical end elements or logical conclusions that follow from any causal state for example if you look at any word like red there are multiple

possibilities possibilities conclusions that can follow from it like you know red blooded or red ruby red meat and so on

so those conclusions the information conclude in those conclusions that are relevant to the end user

so that will be the semantic information so we actually analytically formulate this from a category theory perspective

where we uh what we did is we form we defined a uh syntax category consisting of causal states as the object

and a semantics category consisting of the conclusion that follow from any causal state

and copricious corresponds to from a category theory perspective corresponds to the uh you know

conclusions or how we yeah conclusions that follow from any causal state in the syntax category and this

logical end elements we can map it or we can consider it as a functor from the syntax category to the

semantic category so that's how we actually formulated this semantic information measure and we also

discussed received semantic information which can be written as a fraction of fraction equal to semantic

similarity of the transmitted information where semantic similarity can be computed by quantifying the

overlap in terms of the copricious of transmitted semantic causal state and reconstructed causal state

yeah so further uh yeah so further we perform certain nashe equilibrium analysis so wherein we first did some
equilibrium analysis for some trivial cases like the cardinality of the representation space is one so that
actually means we transmit same semantic symbol for all causal state so that is minimalistic but you know receiver
may need not be able to reconstruct the meaning with high fidelity and we also consider the separating
equilibrium where cardinality of u is same as cardinality of w so that leads to high fidelity of
meaning reconstruction but it leads to maximal transmission so another thing is uh the partial pooling
case where the cardinality of u is less than the cardinality of w so that leads to a minimal representation at
the same time to ensure a high fidelity sufficiently high fidelity at the receiver state so in this
partial pooling case so some interesting insights we got so so we actually um showed that the the
partitioning of the syntactic space occur in such a way that we communicate uk for all causal state that
lead to the same semantics at the receiver side so this partitioning of the semantics the causal state
corresponds to a partitioning in the semantic space unlike a partitioning in the euclidean space for
traditional communication systems so that's an interesting insight that we got from this
natural equilibrium analysis so further i conclude this results by you know discussing the summary of key
analytical results as i mentioned we formulated novel semantic information using category theory
and we also derived of upper low upper and lower bounds of average amount of bits communicated in
an es emerging semantic communication system and showed that these upper and lower bounds are less
than that of a classical system showing minimalism and we also derived semantic error probability measures
so that showed that the lower bound on semantic error probability is always less than or equal to the
lower bound on the probability of bitter measure achieved using classical communication system so
what that tells us is that so even if the probability of bitter is maybe high still the receiver may be able
to reconstruct the semantics with high fidelity as possible so that means we can uh it's okay to know
we can avoid these retransmissions as in classical system and we can maybe transmit at a lower rate so that
means that we must transition to semantic performance measure if we are to make efficient use of communication
resources so that's another insight that followed from these results so further i would like to discuss
some simulation results so we for the simulation we actually look at a visual question and answering based
data set where a teacher communicates certain images encoded versions of the images and then uh the
apprentice's objective is to answer certain questions based on these images so we evaluated this uh the
performance of the emergent language uh for different tasks where we started with a simpler task where the
image only contains maybe one object so images here contains objects of different shape color and size
and we increase the complexity of this task so as time progresses what we saw is that uh the amount of
communication rounds required for this emergent language training reduces close to zero so that tells us that
the system is the system is the semantic communication system is gaining more knowledge of receiver uh the
apprentice is gaining more knowledge as time progresses uh so then you know because of this more knowledge it

can actually avoid retraining for the emergent language so that yeah that way we can get rid of a lot of
communication overhead uh yeah so that's the generalizability aspect and we also evaluated the average amount of
bits communicated over the wireless network wherein we showed that for our proposed system we almost
gain around 100x gain compared to classic traditional semantic communication system that does not incorporate
reasoning into their uh architecture and compared to classical system yeah even much more gain is obtained so
that actually shows shows the minimalism aspect that i was uh discussing so that actually uh concludes the
yeah work on emergent semantic communication and next i will briefly discuss um one follow-up work that we did
where we utilize this causal semantic communication framework for a digital twin driven wireless network uh yeah so this got
accepted at a triple general on selected areas in information theory so here one application is as is shown here
let's say we want to control the operations of certain autonomous robots in an industrial environment
uh yeah using the communication from a base station or an edge server so herein uh we assume that okay given that this
particular industrial environment will be a lot of sensors will be deployed in this environment so using the
sensitive information the base station actually construct a uh digital twin of the industrial environment and
uses this digital twin uh it actually you know using the digital twin it actually computes optimal control
actions uh for these autonomous robots in the system so this is within the framework of semantic
communication system and that's what i will be discussing here so yeah so so here this semantic
communication framework again we consider for a two user so point-to-point system wherein the transmitter
it's called an expert agent because it has access to a lot of sensing information and using the sensed
information it creates a world model about the wireless environment or autonomous agent environment
so basically extract the semantic content elements and you know generate the causal graph and using the
causal graph it actually computes an expert policy control policy and others and this okay instead of
commute computing the control policy the transmitter actually computes a causal semantic representation of the
causal graph and the receiver receiver also has a world model at its end but it creates this world model from the limited
communicated information and then it also computes an immediate receiver here is called an imitator because it tries to compute a policy
uh and the objective is that this policy should be as close to the export policy at the transmitter as possible and this is achieved using the framework of imitation learning
and that's why it's called an imitator agent so that's the yeah so this policy imitator policy should be such that
it is quite close to the optimal export policy yeah so as to ensure that the semantic effectiveness of the task at the
receiver side is as high as possible so that's the system model so compared to our previous work here in we

uh went one step beyond in terms of defining the info i mean semantic information so in traditional
mutual information frameworks
it does not distinguish between causal i mean uh directionality of information like
but i of x comma y is basically same as i of y comma x so yeah so but in
in in a causal system we must be able to distinguish between uh i of x comma y if x causes y uh then
yeah i of x comma y should be different when you know uh when it should be different than i of y comma x
so this is achieved through the principle of intrinsic information so given okay let's say assume that
s i t is the state of the system at time t and ic of s i t minus one given s i t tells us about the
course information about the course state which is s t minus one provided by the current state s t similarly
we can define the effect information which is basically the information about the future effect states
so this can be defined using a distance measure between these two probability distribution which
is probability of s i t minus one given s i t and property of s i t minus one similarly for the effect
information and we choose was sustained distance here because of this unique properties that you know
the distance is uh symmetric here so that's the reason why we actually uh resorted to was sustained
distance here and further we can define the concept of integrated information so let's say that this s s i t
is composed of distinct semantic content elements so okay so our objective is to basically perform
abstractions of this semantic content elements by you know filtering out irrelevant states so we can how we
define integrated information for a particular state s i t is that integrated information is the information
conveyed conveyed by uh the states beyond that of what is conveyed by the individual semantic content
elements so that actually means that only if okay let's say the semantic content elements are all
independent random variables then there there won't be any interactions between these different semantic
content elements so that that means that there are some cause and effect
relations between these semantic content elements so yeah we so those set of states will form a
semantic concept with non-zero integrated information so that's how we can actually
uh you know define a semantic concept or extract a semantic or identify a semantic concept from the data so that's one
uh you know yeah um step in terms of advancing the theory of semantic information um that we actually propose in this work
so further i would also like to quickly discuss uh the methodology that we used for so as i mentioned here
we have to transmitter we have to transmitter has to learn the causal graph uh so which is achieved through causal discovery
and also this causal graph should be represented semantically it should have a semantic representation
which is actually represented as z t here and then yeah so then then the transmitter also has to compute a
action control policy which is pi export of action given the state uh future state current state as well as
the set of sensed variables and then it also has to come up with this transition distribution which is
represented as a causal graph so this okay given that we don't have a um yeah we don't we do not have uh i
i mean we do not have information of the true uh posterior distribution of this causal state transition we
actually resolve to uh using a variational biation optimization approach here where we compute an

approximate distribution by minimizing the kl divergence which actually can be formulated as maximizing the evidence lower bound so that's this g expression here and that's how we actually formulated the
uh you know causal graph computation problem and also at the transmit side there is also the problem of computing the semantic states that is achieved as so that is formulated as maximizing the integrated information conveyed by the causal state about the uh no yt which is the corresponding to the inference variable at the receiver side subject to constraints on the bandwidth so how much information this
is that t conveys about s t should be less than or equal to some threshold so that actually tells
about the bandwidth constraint so there are two different problems here so we solve these two
problems by formulating this uh causal graph uh as well as this yeah causal graph as well as the semantic representation using neural generated generative a based neural networks wherein specifically causal for causal graph we used a graph neural network concept and for semantic state representation um for semantic state representation uh we actually used variational autoencoder based uh framework to represent that
and we actually solve this tool two level optimization problem by a nested optimization uh using gradient descent so
that's how we actually solve the parameters of the neural network i won't go into the details of uh yeah details of the problem uh solution so instead i would like to further state certain simulation results that we had so first is uh what we showed through this generative a based uh framework is that or the weld model based framework as i discuss here is that uh we actually showed that compared to classical
maximum likelihood based transition modeling our proposed causal semantic communication system
uh can better represent the network state uh for all yeah for uh diverse snr uh ranges computer classical community classical systems and also we showed that we showed the generalizability by considering that okay
so we started with some particular uh network or autonomous agent environment then after around let's say 6k samples we
actually um changed the environment dynamics then we actually saw that the classical systems will take
you know the traditional system which are basically data driven ai based algorithms like
i mean transformer based encoder or decoder that takes large amount of time to actually record back to the normal state but our causality basis system is able to adapt quickly to this uh network dynamics so that's the that actually proves the dynamic adaptability aspect uh yeah so that actually concludes my work on causal semantic communication for uh digital twin based uh systems so further i would like to quickly discuss
some future perspectives uh so as i mentioned uh currently i've been considering only a point-to-point
um to make this is

and control system so herein we can consider that each of these user has access to only a
partial observation so basically a case of partial observable markov decision process so
that way we can actually formulate that so using this partial observation and also using the
communicated information from other users each user should form a world model about the environment
yeah then the quality of the using the world model that they construct they come up with a shared
semantic language or shared symbol system yeah so that actually will depend upon the quality of the
world model learned by each of the users so here a promising approach is to actually use the
perception and action planning framework design a perception and action planning framework through
active inference concept so in this direction uh yeah so i am actually looking at uh formulating
the semantic encoder or also representation frameworks are as follows so let's say we have a semantic
encoder at semantic encoder uh so that is defined as an approximate distribution which is q of
ut where u is the semantic symbol and is the t k correspond to the causal states uh pursued at
different uses where k corresponds to user index and then causal representation at each user we can
actually consider it as q of is the t k given is the t minus one ut minus one and here o or t k
corresponds to the partial observation at user k y t minus one k corresponds to the inference decision
in the previous time slot uh at for for user k so here observation basically it's like composed of two
aspects one is like the partial observation and also like the inference decision that this particular
user will uh make yeah so so basically the causal representation or the the world the world model actually
changes uh based on the inference decisions of each user uh and yeah semantic symbol prior is prior is p of ut
and then the logical reasoning and observation model we can actually formulate it like uh probability of yt
plus one given is the t times probability of ot given by t is the t so observation depends on
inference decision as well as the causal state and then the actual transition model will be
as is shown here uh yeah so here um yeah so actually motivated by the formulations in this
uh recent paper um that appeared in rk we can actually formulate uh the collective uh free energy based
uh yeah we can actually write the collective free energy for this particular um yeah but a particular
multi-user semantic communication system um so yeah so wherein i'm not going to the details of this
mathematical expression but we can um write this free energy theme as consisting of three different
factors so one is the reconstruction aspect so how much the user can actually reconstruct the observations
based on the causal world model and then the second time is actually corresponding to the
um causal dynamics approximation which means how much the user use each user can approximate the causal
um world model based on the observations partial observation at its end
um and the final factor will be corresponding to the uh a time that actually regularizes the entropy of
semantic engorder which actually means that so this this time would be actually corresponding to the
um to minimizing certain communication cost so or basically uh ensuring that we communicate minimal
semantic semantics from the data yeah so we can actually formulate uh two um phase
yeah like an active inference we can actually uh formulate the free energy expressions for

the perception phase and also for the inference phase which is corresponding to yeah where y corresponds
to the inference variable and yeah so then the objective is basically um using the uh yeah the free energy for
the perception part we can actually compute the minimalistic semantic encoder or the common semantic
system system as well as using the inference uh free energy for the inference uh free energy for
the inference uh we can actually compute the best inference decision that each user should make
using its own causal uh state perception as well as the communicated information so that's the
you know that's the framework uh that i think can be applied for this multi-user semantic
communication systems uh yeah so that will yeah with that i will conclude my talk so as i uh mentioned um 5g could
not meet all the promises um however as we move towards next generation system wherein uh our yeah so the
communication networks are not just about building better um um wireless i mean better um standard i mean better
systems which high data rate or low latency and so on but also incorporating aspects like sensing or
trustworthiness or and also autonomous control of uh distant agents or cyber physical system so we should
transition to a telecom brain based vision which is consist of consisting of modules like sensing perception
world model and action planning and i discuss in detail few of our work on semantic communication where
we actually showed that using causality and neurosymbolic ai we can actually build minimalistic generalizable and
high fidelity semantic system so they're saving a lot of the bandwidth uh and then yeah the future research
vision uh my future research vision is centered around building sustainable trustworthy and socially
responsible next ngai native wireless systems using the stelacom principles yeah so with that i will
conclude my talk and if you have any questions i am uh glad to answer those thank you for the awesome
presentation okay i will prepare my questions and also welcome anyone in the chat to write and just while
i crop and prepare that just could you share a little bit how did you come to be setting up the lab like
just what was your trajectory from through school on through here how did you get into this research
area uh yeah so i did my phd in um you know 50 generation wireless communications with focus on the
physical layer aspects like you know building uh developing uh beamforming solutions or channel estimation
schemes using conventional signal processing algorithms like you know specifically even approximate
bias in inference kind of techniques um uh yeah so those are the those are my main focus during my phd
but when i transition to a postdoctoral position i choose a different path like i mean i i yeah i saw that you
know uh the physical layer advancements is kind of you know the physical layer literature is kind of
saturating out so we need some new revolution uh in communication system as we move towards next
generation systems that that can yeah that can be um feasible by the incorporation of ai but not just
you know data-driven ai but um a that can actually reason like human beings so that's so that is the motivation
where we are starting to work with uh professor valid sad on um this particular topic um yeah so
that's how i actually entered into you know semantic communications uh which is actually a revolutionary
approach that uh that is aimed at build and improving the resource efficiency of future communication

systems um so yeah so that's that's how i ended up at this uh this particular research
is um is known to make this is known to make this is known to make this is known to make this

excited to see where this all heads because it's aligned with many of my interests and I also believe it's a critical infrastructure kind of civilizational technology layer. So first I tried to summarize in one sentence because there are so many pieces and so many formalisms that might be new. So my one sentence summary was dynamics and constraints of semantic transfer in hybrid and synthetic network intelligence systems. And then to kind of expand on that a little bit so I'll just make a remark about where I saw active inference coming into play and some of the important areas. So starting with a dyadic model, templated dyadic communication models like teacher-apprentice or adversarial models, you are looking towards using active inference to allow real-time cooperation and turn-taking in multi-agent systems. And that would allow the embodiments, which is achieved in different engineering solutions in different ways. However, the top-down approach is what is the analytical formalism that gives a unity from a modeling perspective to all of these different engineering approaches. And those formalisms that you presented on allow the embodiment of agents or situation, class, instance, and context-specific cognitive mechanisms and dynamics, which broadly correspond to perception on the inbound and action selection and on the outbound with reasoning happening in between. And so what that is heading towards is a first principles approach that applies to semantic dynamics, which could be used within a single-threaded shared database state multi-agent system. So it could be a system with a very specific logistical order or known computational resource constraints all the way on through nested arbitrary systems. And then importantly, embodied cyber-physical networks where there can be the explicit introduction of physical and energy models, bandwidth and resource costs. And so that brings a complementary view from the engineering side on informational and cognitive constraints with important physical constraints of the ever-changing basis of networks.

So I just kind of wanted to echo back because there are so many important pieces in the talk and it was a lot of information and it's a part of our tech stack that just kind of works. Like the phone just has service and it's really abstracted.

And then of course we know that the phones are coming out with different versions and I'm not knowledgeable in these areas that much, but it's like services different in different regions and things change and technology happens like that.

So giving that kind of view into the tech stack is also, I think really critical.

So usually I do not give a sort of extended comment, but there's just, I was writing down so many notes. So do you have any thoughts or ideas on that?

Yeah, I think you have described, I mean, the summary of this token, yeah, quite well. I mean, that's a great summary. So yeah. So one point just to conclude is that like, you know, so right now with 5G, we are able to get, you know, large download speeds or lower latency, you know, streaming and so on.

So, but you know, but when we actually move to next generation system, it's not just about cellular communication. So network should also, you know, we should be able to build much more smarter environments, like, you know, smarter transportation systems or smarter, you know, industries, telehealth systems.

So that actually requires the network to actually have a perception about the physical world.

So, you know, if the network has to, you know, control the operations of these distant autonomous agents.

So, yeah, so the network should have a accurate perception about the physical world.

Unlike in classical system where, you know, we just treat the communication systems just like a passive system. So whatever message comes in, you just, objective is just to communicate with the physical world.

So that's the space just to communicate that messages with, you know, high accuracy at the receiving end.

So that's, so that's a transitional paradigm that, you know, we are looking at as we move towards 6G or, yeah, 6G and beyond.

Yeah, another area I wanted to remark on and part of that transformation is predictive processing and the role of prediction in compression.

So the predictive processing frame differencing is used in standard syntax information driven compressive systems.

Like frame differencing in the video, if all you have to do is transmit what changes between frames within a given local minima of minimal compression.

And that was discovered in the computer science area and then actually entered into this broader discussion about like visual and sensory systems.

So that's using frame differencing to, for example, transmit a 4K video with less syntactic information flow.

And then actually it goes even further with all of your work exploring the transfer in a latent or semantic space, which could be more like a primary lexical semantics or conceptual modularity, all the way up on through possibly complex pragmatics like action directives or complex information being shared at a higher semantic level.

And then that's kind of, again, where active inference comes into play is having a unified framework that on one hand can give an account for why frame differencing at a purely sensory kind of Shannon information level is a compressive strategy.

And then use on a latent space in exactly analogous mechanism to enable semantic compression.

So like if we both have a friend ABC, but we both know them as A, then there's the question of the bits per second of the audio transfer.

Like how good is my microphone and how much data needs to be sent?

But maybe a given bandwidth, you only need to say A because B and C are semantically known.

And so that just kind of shows like it's not just about the amount of data transmitted.

Compressing it, all other things being equal is good.

Why send extra data through the wire?

So what kind of information can be transferred?

Transmitting vector embeddings have a markedly different semantic impact than, for example, dialogue between LLMs.

Like the kinds of bandwidth that are currently being used in natural language-mediated transmission.

Could strategies like multimodal or semantic compression give some kind of order of magnitude modifications in the bandwidth or the impact of communication amongst latent space?

Yeah, so one way to look at this, like, okay, so let's say if you have like two LLM agents communicating.

And let's say, okay, so they are using this natural language to communicate among them.

So, yeah, so, yeah, so, yeah, so, yeah, so, so I, let's, let's say, okay, so in, in semantic communication, so we are, I mean, so, so we are not really concerned about, you know, how the, how LLM agent one communicates in a, you know, grammatically correct English sentence.

So, you know, we will be, you know, grammatically correct English sentence.

So, we are,

each agent has um like you know llm agent one so okay so when when we when two humans actually start communicating okay when when two new uh um yeah so when uh one when i start communicating to another person whom i don't know before so initially there is a lot of overhead like you know we have to introduce ourselves we have to share some background knowledge about ourselves um so then further as they become friends so then when they communicate so they they have a lot of you know implicit assumptions about um the other person how that person would be thinking or how that person would be um weaving the world around him so the communication would actually be more um you know more optimized for optimized considering all also that background right so similar to that so yeah so if two llms agent two llm agents communicate among them so so using an emerging language framework our objective is to build a machine language that is sufficient enough enough for llm agent to pursue or understand what llm agent one actually means so that is so that kind of a machine language construction uh leads to a better usage of you know communication resources um so does yeah does that make sense to you yes like where that kind of takes me is when we're talking about emergent semantics and pragmatics which could be between family members it could be between strangers humans it could be online or in person it could be two llms it could be a person or lm like every kind of tony um having this approach gives a way to scaffold the informational dynamics and kind of separate out the syntactic layer from the semantic layer and then it naturally leads to these kinds of sustained duration semantic learning strategies like if every single message was coming from a different source, then there would be some sort of like no free lunch, but you could take the best semantics as possible to each fresh source, like if it was just a different every single time. But then to the extent that you can have a prior about a given type of interaction all the way on through the most specific prior, which is exact relational with a given other entity, then that allows semantic communication strategies like one means this dinner and two means that dinner. Well, then they only have to send one letter. And then when we get into these situations where it can be machine-machine semantics, then that's kind of

what tokens are. Like that's why words in a given language get truncated up into these essentially probabilistically determined semantic units. Like, oh, it's this verb and then ed to be the past tense.
So it's like, or whatever the tokenization strategy happens to be. And then how do you use your tokens to have this no regret or minimal regret, best possible token conveyance linearity to finesse the translation of elements of a semantic space, which could be static or dynamic, into a linearized sequence of messages.
And then the network perspective is really important because network properties can matter.
Like if you have a network where 100% of packets are delivered, you're going to have certain kinds of strategic constraints in communication. Whereas if you have a network where a given percent are being at this, you know, redirected or this kind of, et cetera, is happening, like those physical constraints do matter for a communication strategy. And so having a sort of integrated analytic approach to modeling explicit syntax constraints, classical Shannon syntax, entropy driven computer science and networking on through abstract cognitive informational constraints, like we're just going to assume that it could be the maximally informative symbol, like the letter distribution in natural language could be made more informative. So using some rebalanced, even A through Z, there could be increased information transfer,
but that would only be in certain situations, like a machine-machine situation might be able to do that kind of hacking into the actual symbols being set in plain text. But that just again reflects on how there's both the particular network and symbol set constraints that are really, really well laid out from a Shannon syntax approach. And then there's this outer space, which we're still just beginning to explore, that is grounded in the primary symbol entropy and the observations. However, it can have multiple layers of cognitive and metacognitive dynamics, which are not so closely bound to the primary Shannon constraints, but still do have important constraints of their own. However, that's what semantic information is about.
Because semantics can be about semantics. I think that was one of the questions on one of the slides, which is like cognition on cognition or metacognition.
Yeah. So it's like a, you know, above this communication constraints that we have, we are effectively forming a semantic layer between the transmitter and receiver, which is actually, you know, constrained by all this cognitivity.
Um, yeah. And then the understanding aspects, right? Yeah.
Yeah. That's very insightful discussion.
Thank you. Yeah. Well, just in sort of closing, what are your next steps or what are like the timeline with the lab or how would people learn more or get involved?
Um, yeah, so, um, yeah, I'll be starting my, uh, new position from August, uh, 2025.
Yeah. And yeah, I'm, uh, looking to recruit one to two PhD students. Um, so yeah. And also now continue this work with the students on this particular topic. And I, yeah, I mean, I do have a, uh, lab website, which is still kind of under construction, but, you know, I can share this. I mean, I will share all my updates through this website. Um, yeah. Awesome. Yeah. Happy to, uh, discuss further or even give a talk later when, when I have more results. Yeah. Cool. Well, again, very appreciative of,

of the work and the concise presentation. Wish you the best.

Thank you very much. Thank you for the invitation. Bye. Bye.

Bye.